Databuses and Architectures

CEC 300: Computing in Aviation and Aerospace

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About this Module

- Focus on how the systems come together to achieve the guidance, navigation, control, and mission execution of an aircraft or spacecraft.

- **Databuses** – ensure the transferability of data between avionic components and vehicle systems.
  - Wires or wireless
  - Topologies include
    - Point-to-point
    - Ring
    - Star
    - Networked

- **Architectures** – Integration of flight control system, mission systems, sensors, actuators, computing, internal communication, external communication, etc.
  - Databuses are a major consideration with an architecture and vice-versa.
  - Aircraft / spacecraft specific
  - Standard architectures

Major Considerations

- Electromagnetic Compatibility (EMC) and Interference (EMI)
- Bandwidth and data rates
- Deterministic behavior
- Signal Integrity
- Reliability
- Fault Tolerance
- Scalability (new/upgrade appointments)
- Quality of Service
- Communication protocols
- Environment
- Miniaturization

- SWAP-C
  - Size
  - Weight
  - Power
  - Computing

Different vehicles, platforms, shapes, and sizes all influence the architecture and its databuses

AVIATION DATABUSES
PART I: INTRODUCTION

How did we get from this?

Cessna 172

And this…

Boeing 707-329

And this…

F 104

To this…

Cessna 172 with Garmin G1000

and this…

Boeing 787

To this…

F 35

Introduction

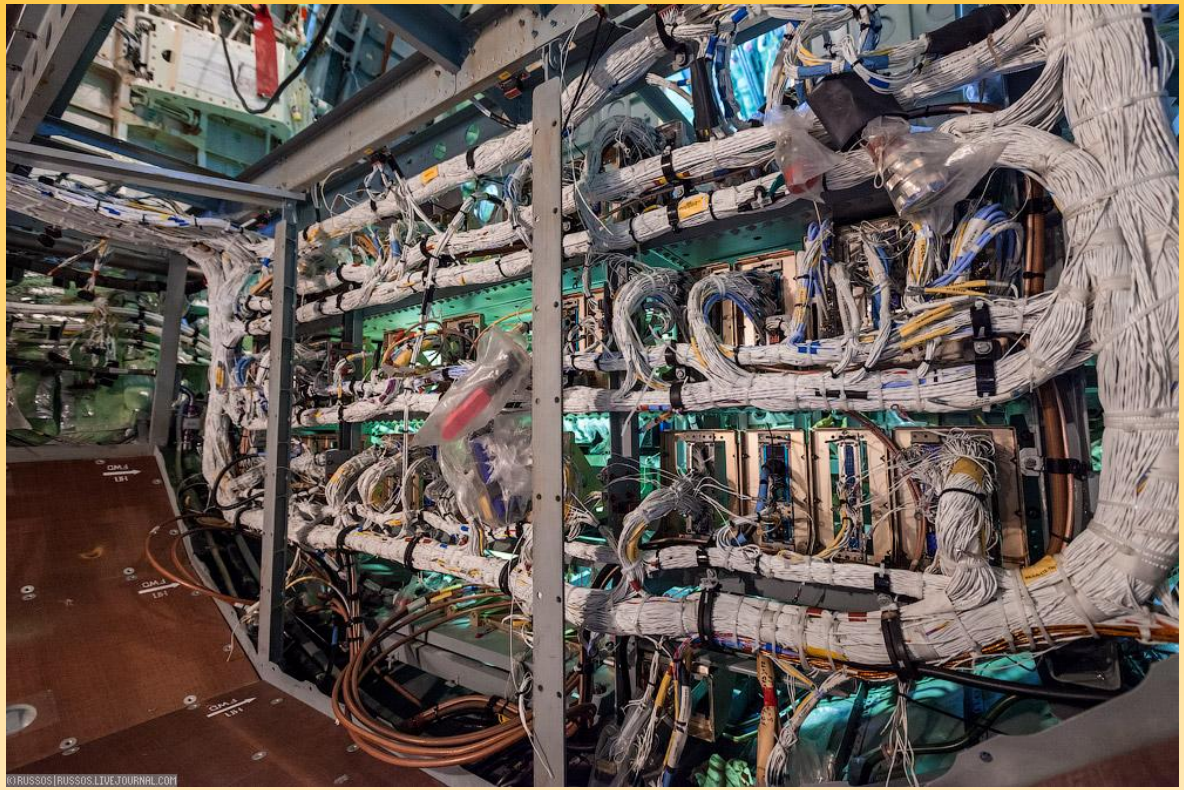
These changes are possible thanks to major advancements in:

- Sensors' accuracy, precision, size, weight, and power.
- Processor and memory capacity, size, weight, and power.
- Display size, weight, and power.
- Evolution from analog to digital electronics.

The data traffic within the aircraft has increase from none to large.

Therefore, **Data Buses** play a big role in modern avionics.

Introduction



Network!

General Time Sensitive Network (TSN) goals:

- Reliable communication over Ethernet
- Ultra-low latency
- Deterministic end-to-end latency
- Secure

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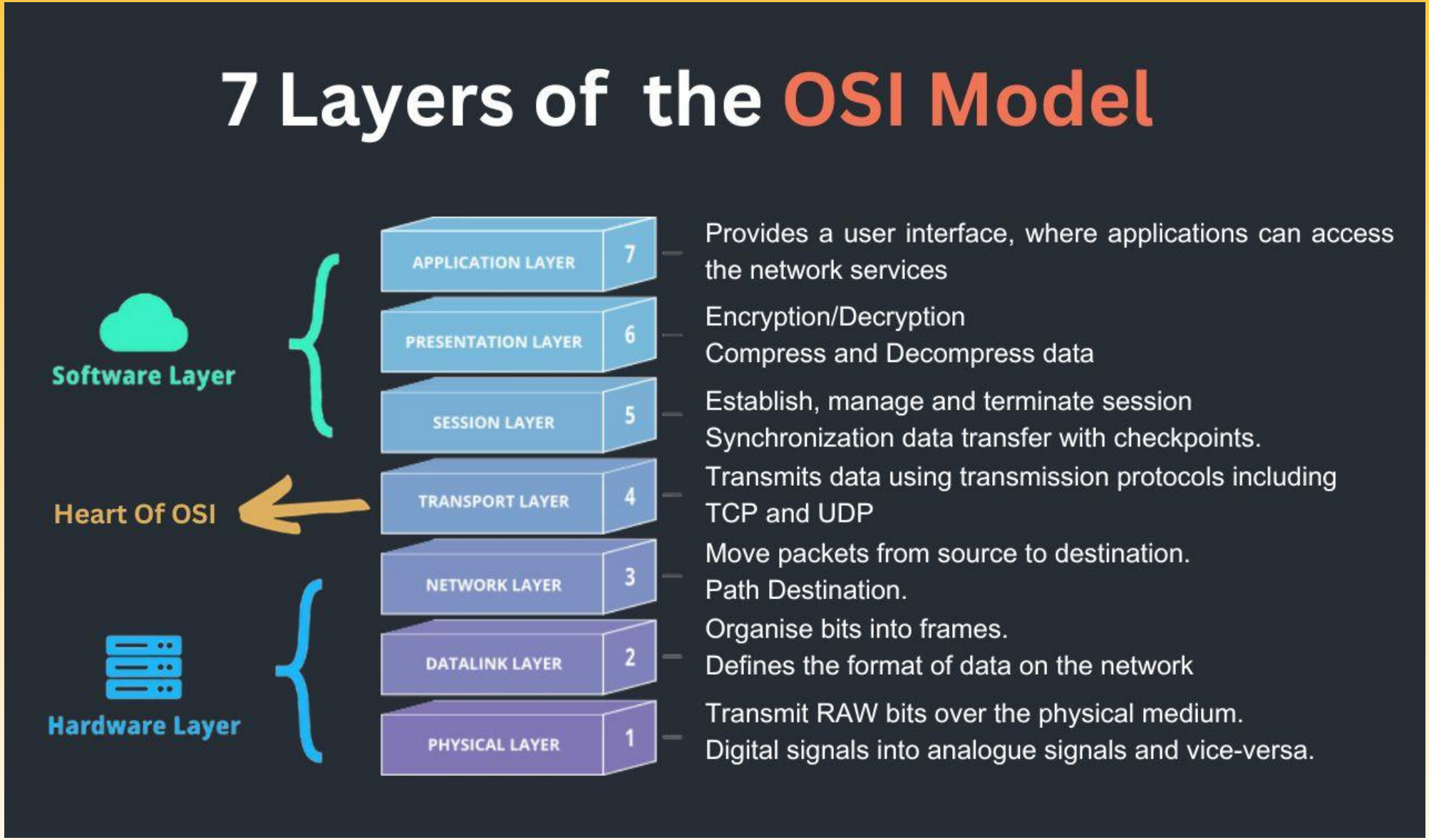
By the End of 90's…

- Aircraft are still using analog signals and some digital busses such as ARINC 419/429, MIL-STD 1553 and ARINC 629.
  - These buses offer low bandwidths (12 kbps -2 Mbps)
  - Extensive wiring is required
  - Limited data formats. Difficulties to support sophisticated communication protocols.

Needs

- **Bandwidth**: the amount of data generated in airplanes have considerably increased in recent years.
- Need for **bidirectional communication** to support complex protocol.
- **Increase the sharing of data**: interdependence of systems is increasing (data bases, clocks).
- **Flexibility** of communication architectures.
- Use of standards protocols of communications:
  - Aircraft are connected with a large network

OSI Network Model



What is OSI Model in Computer Network | 7 Layers Explained - CodeRepublics

Layer Types

- **Application Layer**
  - Requests issued to presentation layer
  - Receives responses to requests from the presentation layer
  - Examples: HTTP, FTP, POP, etc.

- **Presentation Layer**
  - Provides a common interface to a variety of different application layer entities.
  - Enables encryption/decryption and compression/decompression

- **Session Layer**
  - Establishes, manages, and terminates sessions
  - Enables synchronization (e.g. streaming)

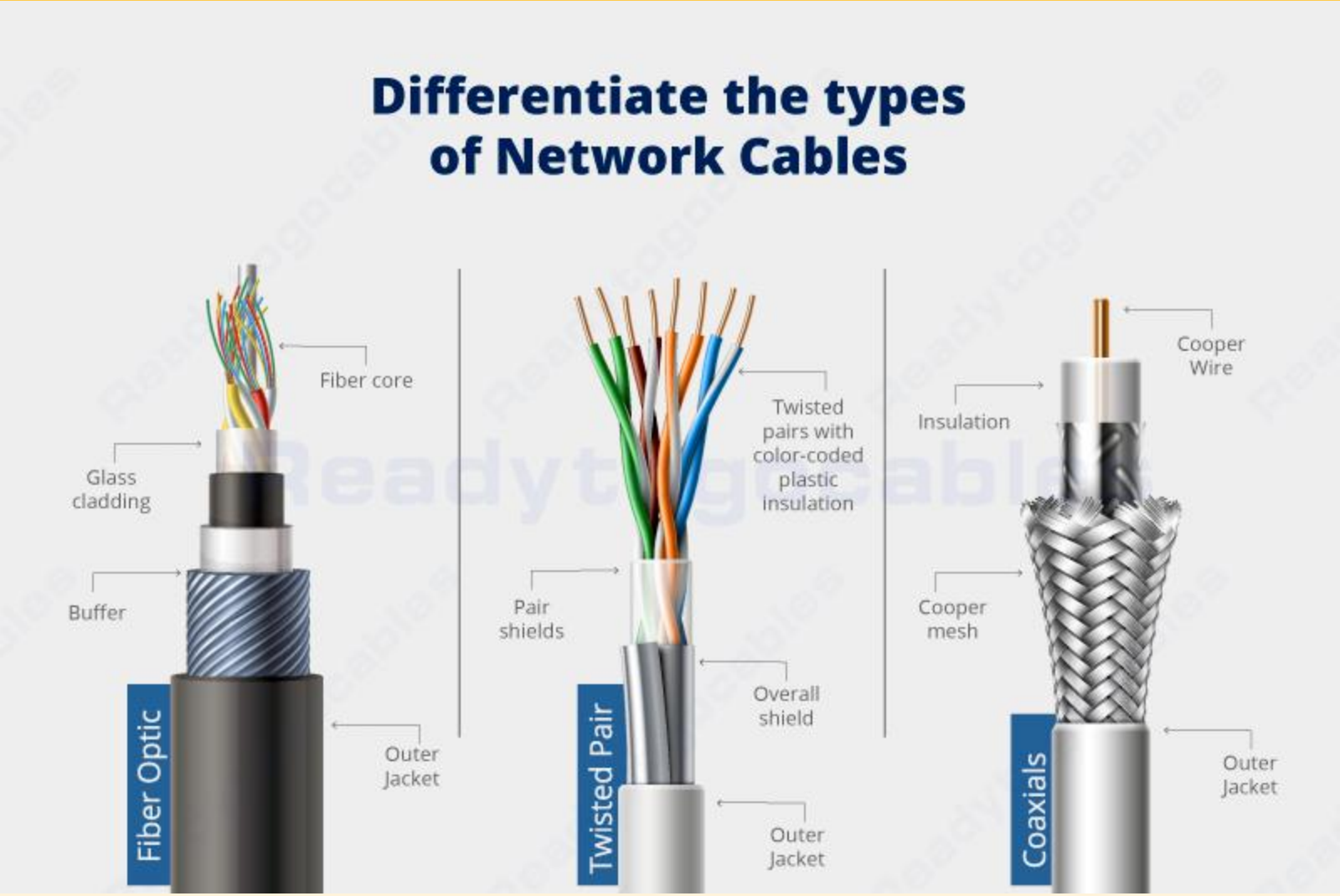
- **Transport Layer**
  - Responsible for the transportation of data packets between entities.
  - Port addressing, segmentation/reassembly, connection management, reliability and error control.
  - e.g. Transportation control protocol (TCP) and User Datagram Protocol (UDP)

- **Network Layer**
  - Constructing data packets
  - Routing
  - Local addressing (e.g. Internet Protocol (IP))
  - Traffic control and congestion management

- **Data Link Layer**
  - Node-to-Node data transfer
  - Organizing raw physical bits into frames
  - Medium Access Control (MAC) addresses for local delivery
  - Error control (e.g. CRC)

- **Physical Layer**
  - Bits to signals / signals to bits
  - Physical medium & topology
  - Data rates
  - Transmission modes
  - Hardware components (Cables, network cards, hubs, repeaters, etc.)

Transmission Lines



How to differentiate the types of network cables? - Readytogocables

Electromagnetic Interference:

- Consider:
  - Line-to-line
  - External-to-line
  - Line-to-external
- Signals can ingress/egress to/from a transmission line
  - If you block one, you block both.
- Mitigations:
  - Twisted-Pair – two transmission lines with the signal on one and its inverse on the other produce B-fields that cancel each other.
  - Shielding
    - Braided/mesh shield
    - If multiple twisted pairs, each pair can be shielded and then the all pairs are shielded together.

Wrapping Up

Last Time:

Learned about the major avionic systems associated with GNC

Today:

Discussed the motivation for the use of Avionics Databuses and Architectures.

Introductory concepts were covered.

Coming up:

Aviation Bus Standards: ARINC 429, ARINC 629, ARINC 664

Questions?

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Module References

- Past Presentations
  - Lecture slides prepared and reused from Dr. Timothy Wilson.
  - Lecture slides prepared and reused from Dr. Eduardo A. Rojas-Nastrucci
- “Principals of Avionic Systems” by Al Helfrick
- “Digital Avionics Handbook” by Spitzer, Ferrell, Ferrell.
- “Avionics Full Duplex Ethernet and the Time Sensitive Networking Standard” by Pasquier and Schneele, Airbus.